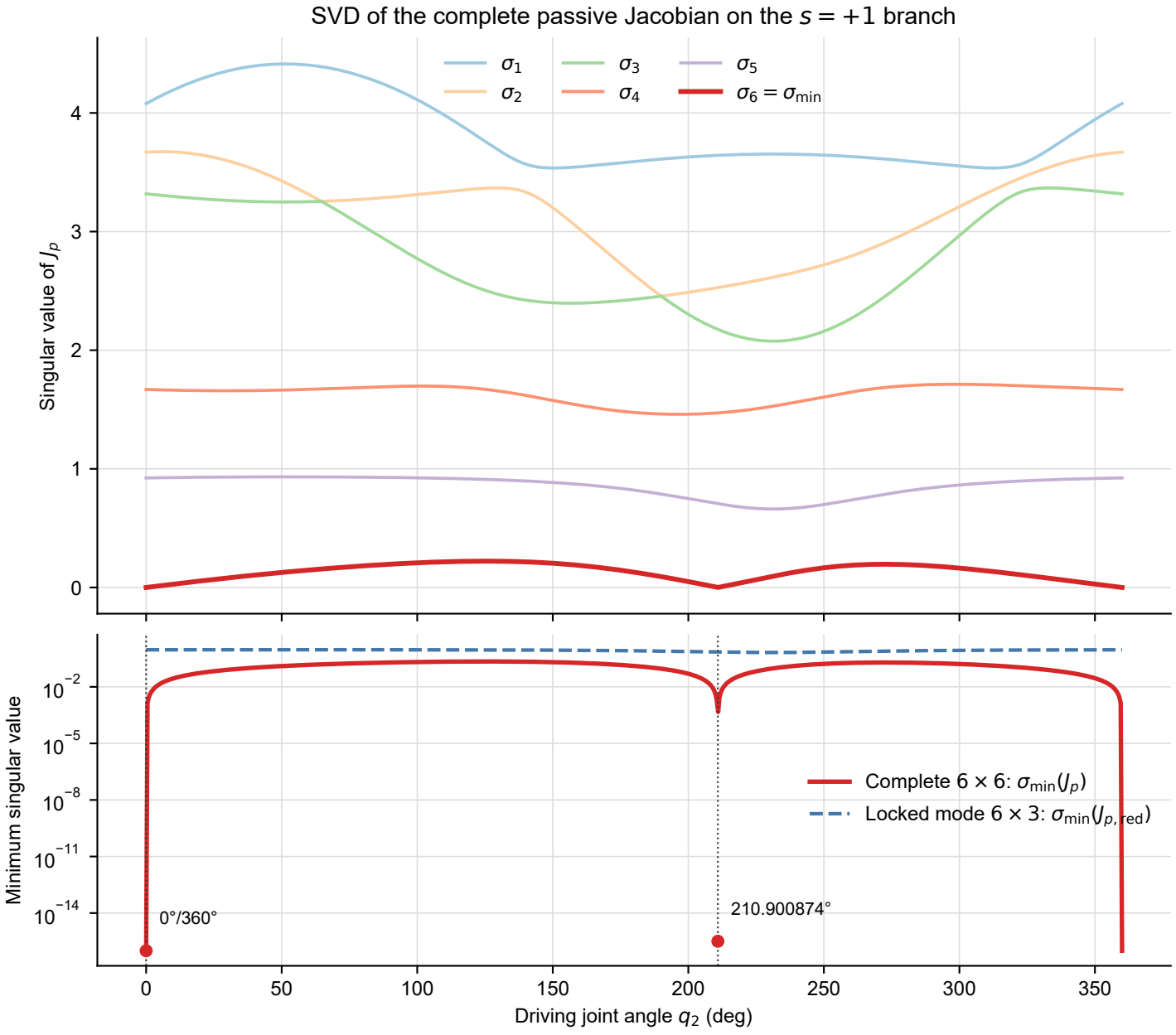
SVD of the complete passive Jacobian on the $s = +1$ branch
σ_1
σ_2
σ_3
σ_4
σ_5
$\sigma_6 = \sigma_{\min}$
Singular value of J_p
0
1
2
3
4
Minimum singular value
10^{-2}
10^{-5}
10^{-8}
10^{-11}
10^{-14}
Complete 6 × 6: $\sigma_{\min}(J_p)$
Locked mode 6 × 3: $\sigma_{\min}(J_{p,\mathrm{red}})$
0°/360°
210.900874°
0
50
100
150
200
250
300
350
Driving joint angle q_2 (deg)